

UTS Cloud Computing Class - Lab Guide

Week 8 Lab: ESXi Install and Host Profiles

This week you will add an additional host into your existing infrastructure, which will require some revision of previous weeks.

Task 1. Prepare for the lab.

1. Open VMware Workstation, and **power on** the following three/four VMs: 'VCSA' + 'OpenWRT' and 'iSCSI' ('DT' for F2F students).
2. Verify that all your machines are online and connected to the network.
3. Connect to your VCSA vm using a **web browser** (from 'DT') (<https://172.20.10.94>).

** **STOP** * Ping from the Terminal (or cmd) window to check if there is a connection to ESXi01 and ESXi02. If connectivity is not established follow the steps outlined from earlier weeks before continuing.*

Task 2. Create a new ESXi host (as a VM).

Follow steps from Week 2: Lab 1: Task 3

From VMware Workstation, create a new virtual machine '**Esxi03**'. Assign the following parameters:

- Guest OS: VMware ESX; version: VMware ESXi 6.x
 - 18GB disk
 - 2 vcpus
 - 4GB ram
 - 4 NICs in VMnet mode
1. File->New Virtual Machine. In the Wizard, select Typical, then set the new virtual machine according to requirements.
 2. Locate the ESXi install iso (**VMware-VMvisor-Installer-7.0.0-15843807.x86_64**) on the class cloud share and attach it to the CD drive of the newly created VM.
 3. Create ESXi in the location of /images/tmp/<student ID>
 4. Select "**Customize Hardware**" to add more NICs. Configure all the NATs to VMNet6 (Hostonly for F2F)
 5. Power up the VM, set password 'VMware1!' for the user root and complete a local install.
 6. Once installed, access the console by press F2 and configure the IP address of the management network.
 7. Login as the root user
 8. Select "Configure Management Network" and "IP Configuration" 9. Choose "static IP" 172.20.10.53/24 Gateway 172.20.10.2
 10. Select "Restart Management Network".
 11. Test the connection of ESXi03. Ping 172.20.10.53 from command prompt (Terminal of DT).

Task 3. Add the host to your existing infrastructure.

1. From your vCenter Web Client connection, add your new host to your existing cluster.
2. Assign a valid vSphere Enterprise Plus license from your pool of available licenses.

3. Put your ESXi03 into maintenance mode.

Task 4. Connect new host ESXi 03 to iscsi.

Follow steps from Week 3: Lab 2: Task 3

1. From browser, ESXi03, select the **Storage** link on the left side panel.
2. Navigate to **Adapters** tab and select 'Software iSCSI'.
3. Scroll down to locate Dynamic targets section and click Add dynamic target. Click to add IP address as '**172.20.20.50**'.
4. Click 'Save Configurations'. Click **Rescan**.
5. Navigate to **Devices** tab and 'FreeNAS iSCSI Disk' should now appear in the list.
6. The previously created 'vmfs' should not be visible on the left panel under 'Storage'.

Task 5. Create Host Profile

Use Host Profiles to create and apply a uniform host configuration. This task is to give you an experience of creating a host template and attaching it to a newly created ESXi server.

1. Power up any one host server ESXi01 or ESXi02.
2. On VCSA web browser, navigate to **Home** by clicking on **vSphere Client** on left top corner of the web page.
3. Look for **Policies & Profiles** option on left side panel. Click **Host Profile -> Extract Host Profile**.
4. Select the powered up existing EXSi host and follow the steps to complete the creation of host profile. Name it *YourName*-Profile.
5. Once completed, again back on the **Home** page, select your third newly created host (ESXi03) from the **Hosts and Clusters** View.
6. Right click on **EXSi03 -> Host Profiles -> Attach Host Profile**. Follow the steps to complete attaching your newly created profile to the ESXi03.
7. Right Click your new host, and from the **Host profile** menu **Check compliance**.
8. From the **Summary**, check if it is compliant with the policy you have issued.

Task 6. Prepare for the next lab.

1. **Power off** your remaining VMs
2. **Shutdown** your other host.
3. **Shutdown** all other VMs in VMware Workstation.